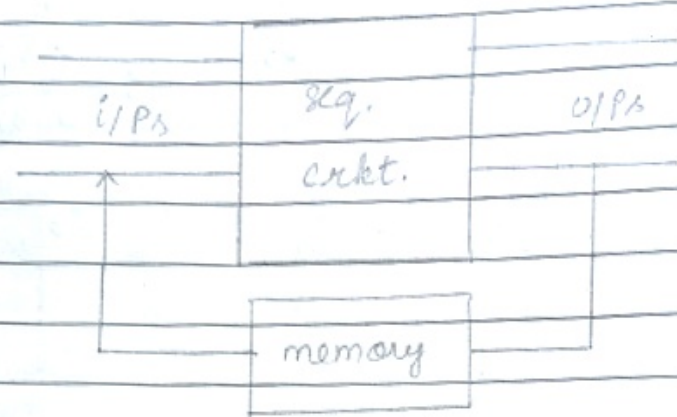


Sequential Circuit

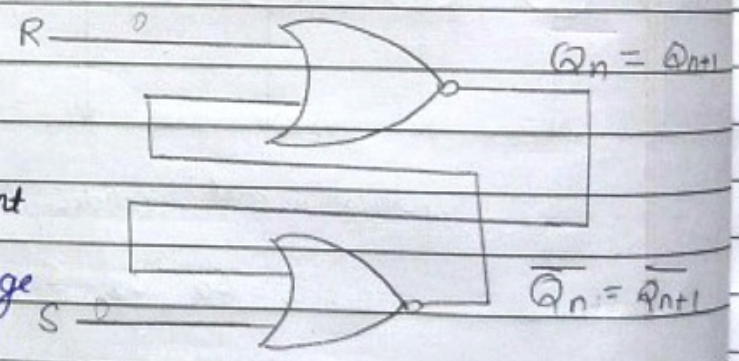


In this, there is a feedback path between outputs and inputs. It means, output depends upon present state as well as past outputs.

Latch

★ R-S Latch

If $Q_n = \bar{Q}_n$, "invalid cond"



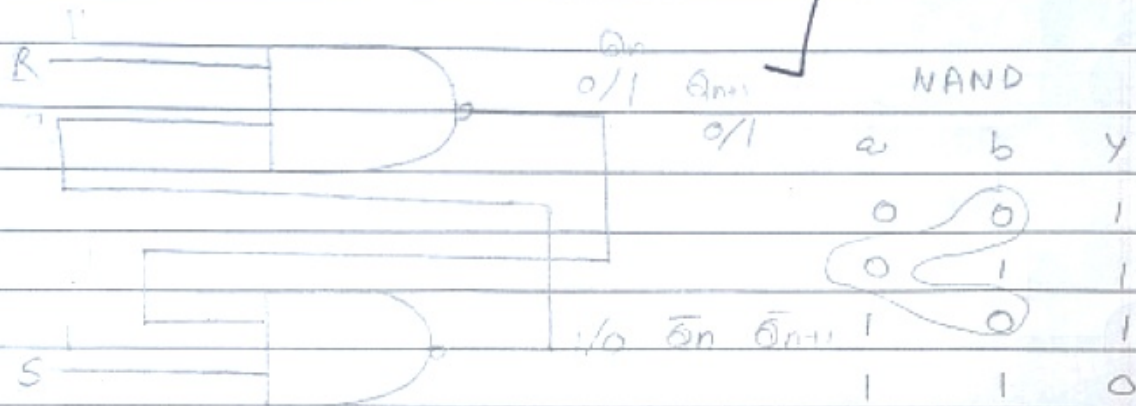
Reset	Set		
R	S	Q_{n+1}	Comment
0	0	Q_n	no change
0	1	1	Set
1	0	0	Reset
1	1	X	Invalid cond"/ indeterminate/ forbidden

#NOTE: If any 1 i/p of NOR gate is high (1), output will be low (0), irrespective of other inputs.

NOR

a	b	y
0	0	1
0	1	0
1	0	0
1	1	0

★ NAND - based S-R / R-S Latch



If any one i/p is low (0), o/p is high (1) otherwise low (0).

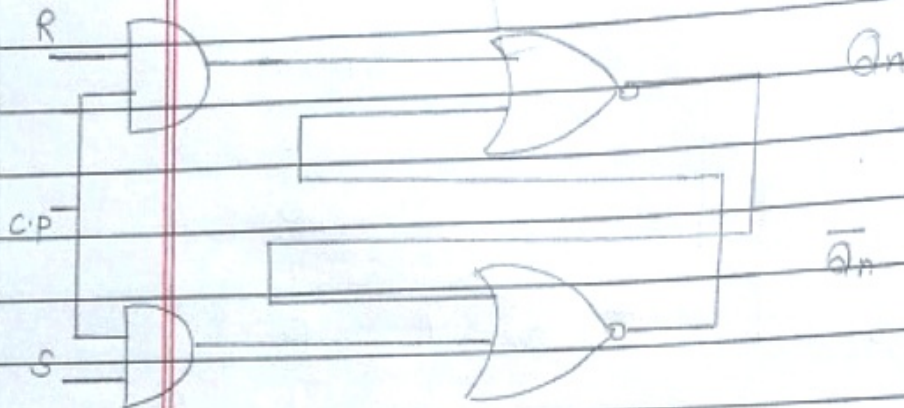
R	S	Q_{n+1}	Comment
0	0	X	Invalid
0	1	1	Set
1	0	0	Reset
1	1	Q_n	no change

★ Flip - Flop

- If we apply a clock pulse (C.P) in latch, it becomes flip-flop.
- Flip-Flop is the smallest unit of memory and it stores one bit at a time.
- It has 2 stable state, that's why it is also called bistable multivibrator.
- It has 2 outputs and both are complement of each other.
- If both are same, this condⁿ is called invalid condition.
- If $Q_{n+1} = 1$ and $\bar{Q}_{n+1} = 0$, it is called set condition.
- If $Q_{n+1} = 0$ and $\bar{Q}_{n+1} = 1$, it's called reset/unset condition.

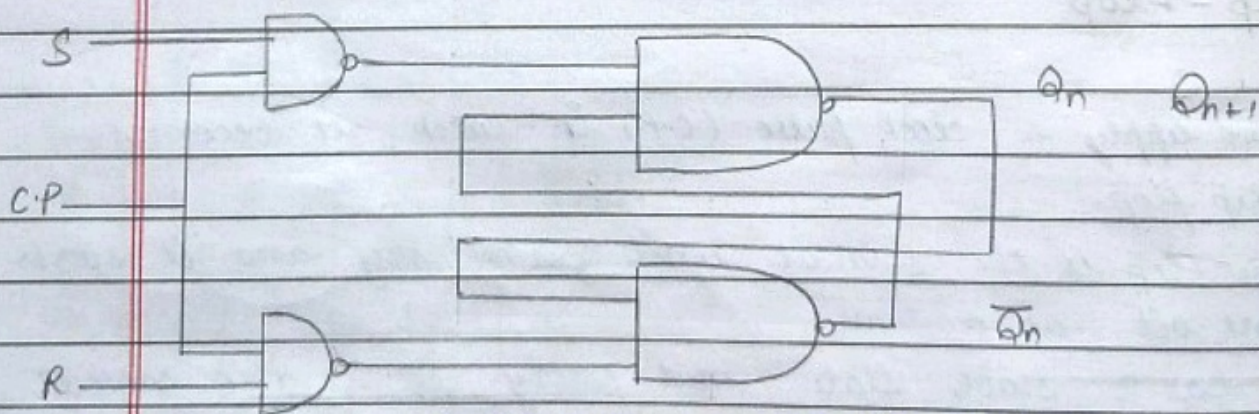
NOR-based

★ R-S Flip-Flop



C.P	R	S	Q_{n+1}	Comments
0	x	x	x	doesn't work
1	0	0	Q_n	no change
1	0	1	1	Set
1	1	0	0	Reset
1	1	1	x	Invalid

★ NAND-based S-R Flip-Flop



$$\overline{a+b} = \overline{a} \cdot \overline{b} \quad (\text{De Morgan's theorem})$$

// interchanged R & S so that table must be same for both types

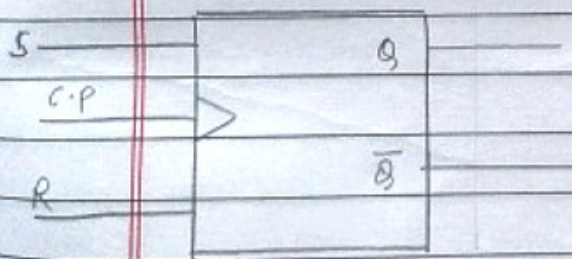
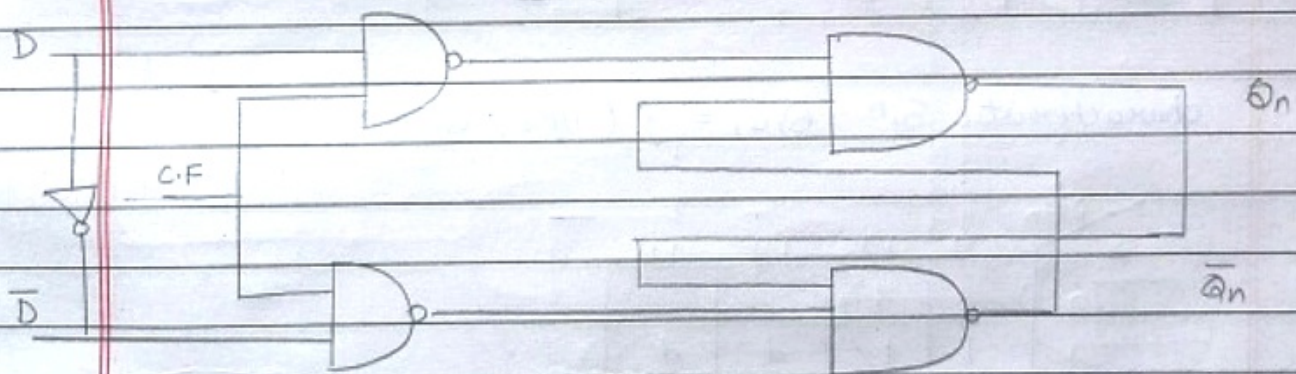
C.P	R	S	Q_{n+1}	Comments
0	0 X	X	X	doesn't work.
1	0	0	Q_n	no change
1	0	1	1	Set
1	1	0	0	Reset
1	1	1	X	Invalid

If any I/P low(0), O/P = 1.

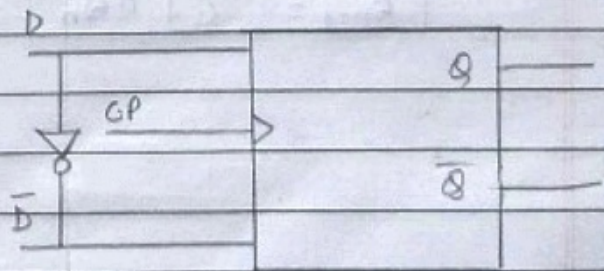
★ DeFlip-Flop

We are getting stable output when both I/Ps are complement of each other. in S-R-Flip-flop.

$$S = D \quad R = \bar{D}$$



Flip-Flop



DeFlip-Flop

$$Q_{n+1} = D$$

C.P	D	Q_{n+1}
1	0	0
1	1	1

- Characteristics Equations of S-R Flip-Flop
From characteristics table

C.P	S	R	Q_n	Q_{n+1}
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	d
1	1	1	1	d

characteristic Eqⁿ $Q_{n+1} = f(\text{inputs}, Q_n)$.

S	RQ_n	00	01	11	10
0			1		
1		1	1	d	d

$$Q_{n+1} = S + RQ_n$$

• Characteristics Eqⁿ of S-R DeFlip-Flop

From table,

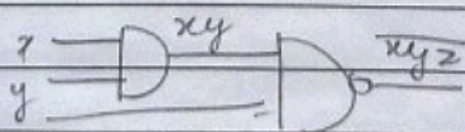
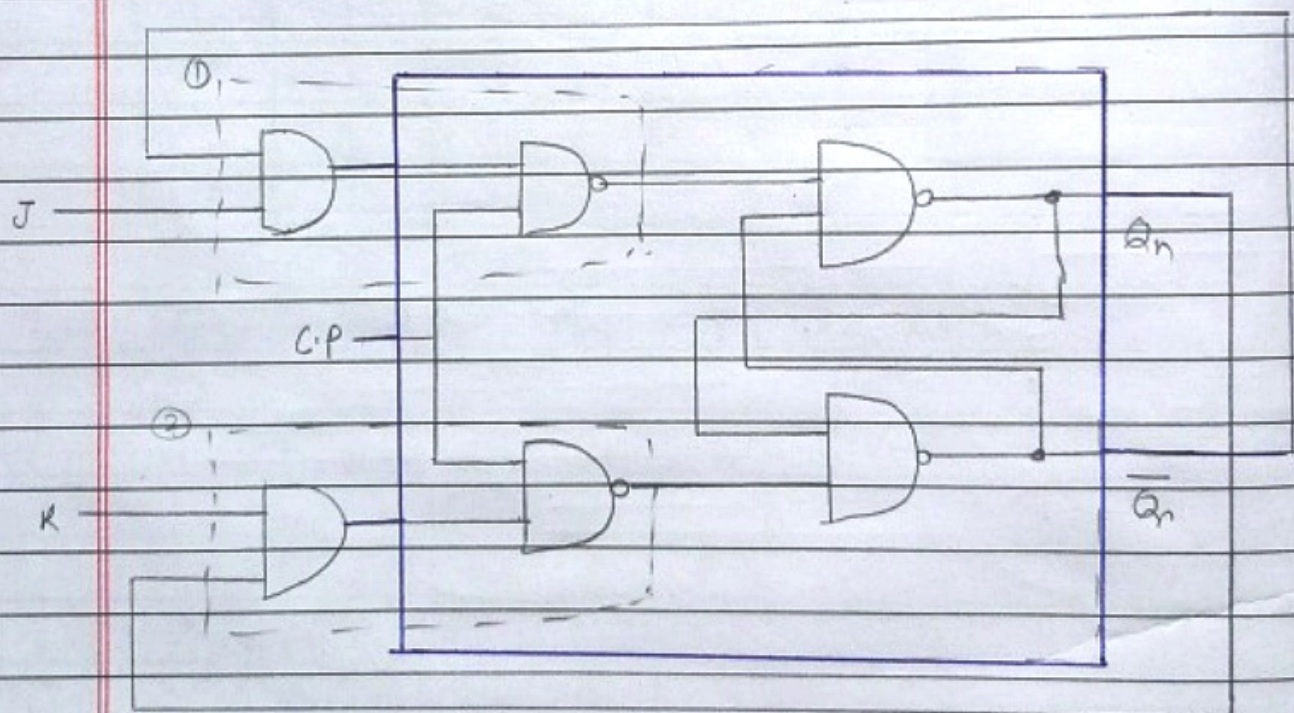
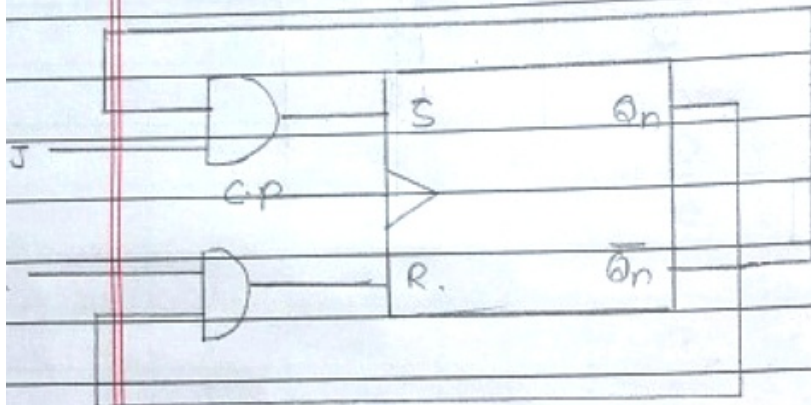
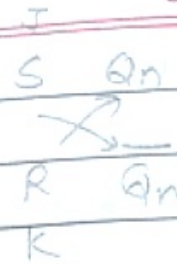
C.P	D	Q_n	Q_{n+1}
1	0	0	0
1	1	1	0
1	1	0	1
1	1	1	1

$$Q_{n+1} = D$$

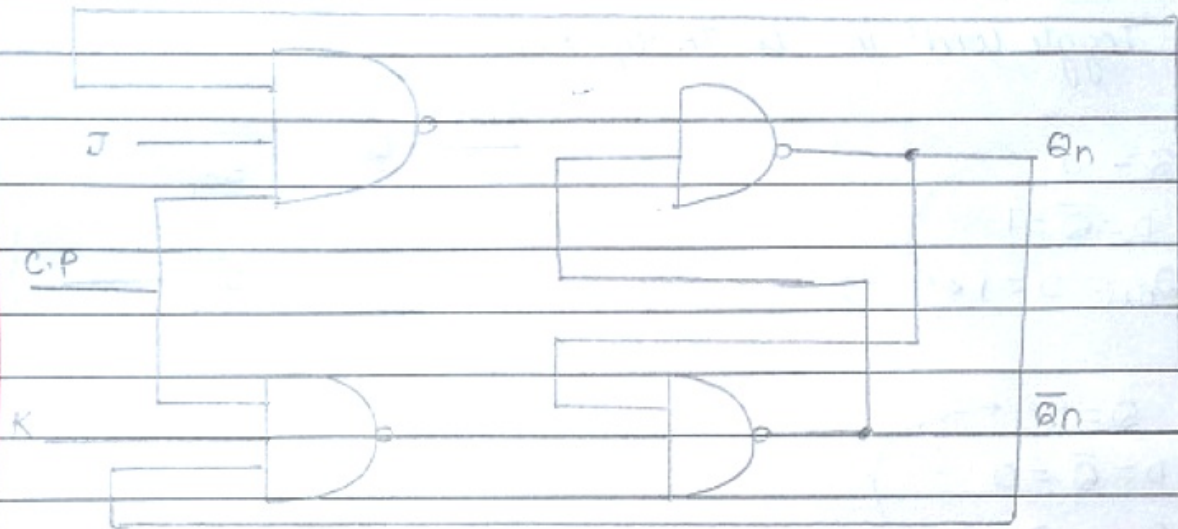
* J-K Flip Flop

$$S = J \overline{Q_n}$$

$$R = K Q_n$$



① & ② can be replaced by 3 i/p NAND gate (7410) IC no.



	C.P	$\downarrow$ S J	$\downarrow$ R K	Q_n	Comments
	0	X	X	Q_n	
	1	0	0	Q_n	No change
	1	0	1	0	Reset
	1	1	0	1	Set
	1	1	1	$\bar{Q}_n$	Toggle (Complement)

	C.P	$\downarrow$ S J	$\downarrow$ R K	Q_n	Q_{n+1}
	1	0	0	0	0
	1	0	0	1	1
	1	0	1	0	0
	1	0	1	1	0
	1	1	0	0	1
	1	1	0	1	1
	1	1	1	0	1
	1	1	1	1	0

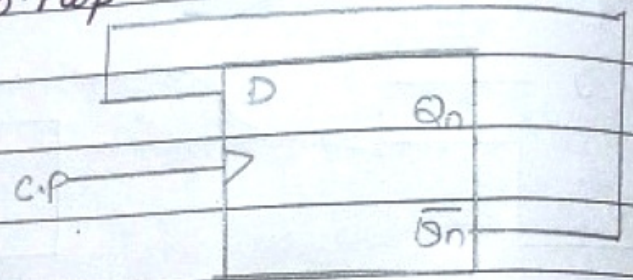
$J \backslash K Q_n$	00	01	11	10
0	0	1	1	1
1	1	1	1	1

$$Q_{n+1} = \bar{K}Q_n + J\bar{Q}_n$$

$$Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n$$

• Toggle Condⁿ in De-JK Flip-Flop

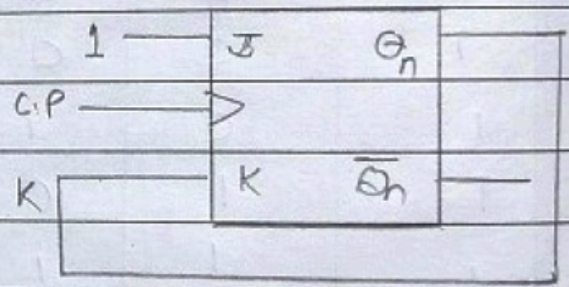
i) $Q = 0$
 $D = \bar{Q} = 1$
 $Q_{n+1} = D = 1$



ii) $Q = 1$
 $D = \bar{Q} = 0$
 $Q_{n+1} = D = 0$

• Toggle Condⁿ in JK Flip-Flop

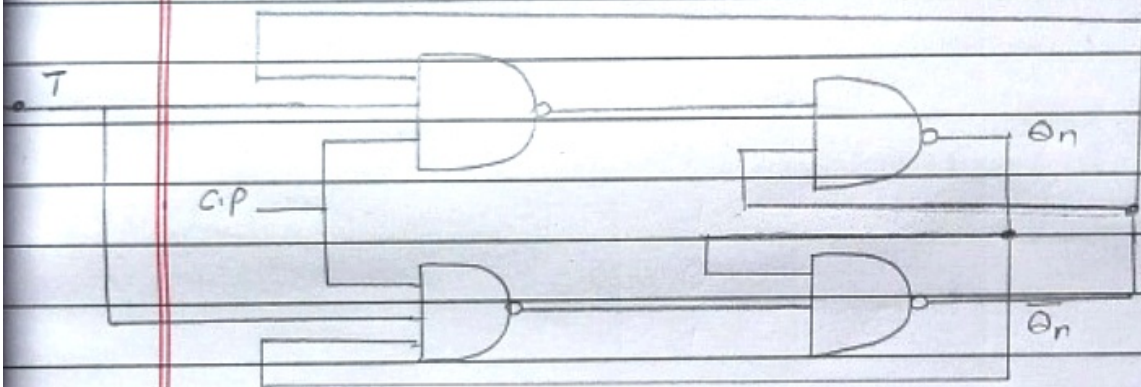
i) $Q = 0$
 $J = 1, K = 0, Q_{n+1} = 1$



ii) $Q = 1$
 $J = 1, K = 1, Q_{n+1} = 0$

* T Flip Flop (Toggle Flip Flop)

If both i/p's of JK Flip Flop made common and connected to T. (i/p), it becomes T flip flop.



C.P	T	Q_n	Q_{n+1}	C.P	T	Q_{n+1}
1	0	0	0	1	0	Q_n
1	0	1	1	1	1	$\bar{Q}_n$
1	1	0	1			
1	1	1	0			

$$Q_{n+1} = T \oplus Q_n$$